PTERO | Thermo RAW to mzML conversion guide

MSConvert settings, MS1/MS2 preservation, validation and reporting

Before you begin

• Keep each original Thermo .raw file unchanged. Install a compatible Windows version of ProteoWizard MSConvert and record its version. Convert each acquisition separately; for multiple samples, add multiple RAW files and generate one mzML file per run.

• Use a dedicated output folder with sufficient disk space. Confirm the instrument/acquisition type and whether the RAW contains profile or centroided spectra.

Recommended settings for a first PTERO conversion

• Input: one or more Thermo .raw files. Output format: mzML (not MGF or mzXML). Choose an explicit output directory and retain identifiable sample filenames.

• Compression: enable zlib for smaller files without intentional loss of numerical precision. Precision: use the software defaults (typically 64-bit m/z and 32-bit intensity), or 64-bit for both if your validated workflow requires it. Record the chosen precision.

• MS levels: retain both MS1 and MS2. For a conventional MS1/MS2 acquisition, either apply no MS-level filter (retains all acquired levels) or explicitly use the filter 'msLevel 1-2' (excludes MS3+). Never use 'msLevel 1' or 'msLevel 2' alone for this workflow.

• Peak picking: if profile data need centroiding for the intended analysis, use the vendor peakPicking algorithm for MS levels 1-2 when available, and place it FIRST in the filter list. If the source data are already centroided, do not centroid them again. For profile-preserving analysis, leave peak picking off. Confirm the centroid/profile requirements of your PTERO workflow.

• Other filters: initially leave threshold, top-N peaks, m/z window, scan-number, charge-state, zero-sample modification, precursor recalculation and similar optional filters OFF. These can remove or alter data and should only be used for a justified, documented analysis protocol.

Step-by-step in the MSConvert graphical interface

• 1. Open MSConvert and add the Thermo .raw file(s).

• 2. Set output directory and choose mzML as the output format.

• 3. Enable zlib compression if desired; record m/z and intensity precision.

• 4. Review the Filters list. If centroiding is needed, add Peak Picking with vendor algorithm and MS levels 1-2 as the first filter. Otherwise leave Peak Picking off.

• 5. To explicitly restrict output to MS1 and MS2, add MS Level with range 1-2 after peak picking; otherwise do not add an MS Level filter.

• 6. Avoid intensity-threshold or peak-count filters for an initial reference conversion.

• 7. Start conversion. Inspect the console/log for warnings and errors and confirm one nonempty mzML output per input file.

How to verify the mzML before importing into PTERO

• Check that both MS1 and MS2 spectra are present (assuming the original acquisition contains both). Inspect spectrum counts by MS level with a compatible mzML viewer or parser and compare against the acquisition record.

• Check representative retention times, precursor m/z values, intensities and scan ranges. If centroiding was applied, inspect several MS1 and MS2 scans to confirm that peak shapes and peak counts are plausible.

• Do not assume a successful file write guarantees correct scientific content. If MS2 is missing, check for an unintended MS-level filter; if peaks appear unexpectedly sparse, check peak-picking and threshold filters.

• In PTERO, select Raw data → Create PTERO Dataset, choose the mzML file path (or the supported directory/multiple-file input workflow), and specify the output dataset location.

Conversion settings to record in a reproducibility report

• Record: sample/run identifier; original RAW filename and instrument; acquisition mode (DDA/DIA, if known); ProteoWizard version; conversion date; output mzML filename; output directory; mzML format; zlib setting; m/z and intensity binary precision.

• Record the exact filters and their order, including peakPicking algorithm and MS levels, MS-level selection, and any additional filter parameters. Note explicitly when no filters were used.

• Record MS1/MS2 counts in the output, conversion warnings/errors, and whether the input was profile or centroided. Keep the conversion log or command line with your project records.

• For future PTERO benchmarking, a structured conversion report can accompany the benchmark results; this reporting is a documentation recommendation, not a currently implemented automatic PTERO feature.

References: https://proteowizard.sourceforge.io/tools/filters.html and https://proteowizard.sourceforge.io/tools/msconvert.html